

Project 2: Thermal-Hydraulic Design and Metallurgical Evaluation of a Seawater Steam Surface Condenser

Sharif University of Technology
Department of Mechanical Engineering

Sina Zerehposh – 401170829

Tina Movasaghi – 401170961

Course Instructor: Dr. Pakzad
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Abstract

In coastal desalination infrastructure and coastal thermal power generating stations, seawater-cooled surface condensers represent critical thermal nodes governing overall plant efficiency and lifecycle generation costs. While copper-based alloys such as Aluminum-Brass have historically dominated seawater tubing applications due to high intrinsic thermal conductivity and biofouling resistance, their susceptibility to erosion-corrosion, sulfide attack, and progressive corrosion-product scaling poses substantial operational burdens.

This project delivers a comprehensive design and comparative metallurgical evaluation for a 24.0 MW steam condenser using **HTRI Xist (TEMA AES)**. A baseline Aluminum-Brass configuration (1.245 mm wall) is systematically contrasted against Titanium Grade 2 in both standard (1.245 mm) and thin-wall (0.559 mm) profiles. Furthermore, conventional TEMA fouling standards are critically reviewed against Heat Exchange Institute (HEI) standards. Finally, an analytical thermal resistance model formulated in MATLAB is evaluated against HTRI results, unveiling a 10.3% underestimation driven by internal tubeside velocity attenuation and convective degradation ($h_i \propto V^{0.8}$).

Contents

1	Introduction and Theoretical Metallurgy Framework	3
1.1	Degradation and Corrosion Mechanisms: Copper Alloys vs. Titanium . .	3
1.2	Fouling Dynamics and Resistance Mitigation	3
1.3	Critique of Standard TEMA Seawater Fouling Tables	3
2	Exchanger Architecture and Parameter Selection	4
3	Thermal-Hydraulic Design & Metallurgical Evaluation	4
3.1	Material Comparison: Base Aluminum-Brass vs. Standard Titanium . .	4
3.2	Optimization via Thin-Wall Titanium ($t_w = 0.559$ mm)	7
4	HTRI Runtime Diagnostics and Vibration Analysis	9
5	Analytical Validation via MATLAB and Root-Cause Analysis	10
5.1	Root-Cause Diagnosis of the 10.3% Underestimation	12
6	Summary and Conclusions	12

1 Introduction and Theoretical Metallurgy Framework

The core task involves the thermal-hydraulic design of a surface condenser condensing 40,198 kg/h of exhaust steam at 154.2°C (500 kPa) using raw seawater entering at 18.0°C and exiting at 42.5°C.

1.1 Degradation and Corrosion Mechanisms: Copper Alloys vs. Titanium

- **Copper-Based Alloys (Aluminum-Brass, Cu-Ni):** Copper alloys rely entirely on a delicate cuprous oxide (Cu_2O) passivating film. Under dynamic seawater turbulence exceeding a critical threshold ($\approx 2.0\text{--}3.0\text{ m/s}$), this protective film shears off, triggering accelerated *erosion-corrosion*. Furthermore, in polluted coastal seawater containing dissolved sulfides, copper oxides convert into non-adherent copper sulfides, inducing catastrophic localized pitting.
- **Titanium Grade 2:** Titanium instantly oxidizes in aqueous media to form a tenacious, self-healing rutile/anatase titanium dioxide (TiO_2) passive layer. This ceramic-grade film resists dynamic fluid velocities beyond 20 m/s, granting complete immunity to pitting, crevice corrosion, and erosion-corrosion in raw seawater across industrial temperature operating windows.

1.2 Fouling Dynamics and Resistance Mitigation

- **Aluminum-Brass:** Cuprous ions (Cu^{2+}) exert a biocidal effect that retards macro-biological attachment (e.g., barnacles, algae). However, progressive substrate corrosion forms heavy, porous scale composed of basic copper salts and carbonates, which drastically impedes conductive heat transport.
- **Titanium:** Titanium lacks biocidal surface activity. However, it exhibits a non-polar, low surface energy state that prevents scale adhesion. Moreover, the superior mechanical integrity of titanium allows continuous high-velocity flow and the deployment of continuous mechanical sponge-ball cleaning systems (e.g., Taprogge systems), sustaining an active *self-cleaning* shear effect.

1.3 Critique of Standard TEMA Seawater Fouling Tables

Standard TEMA tables tabulate seawater fouling factors solely as functions of bulk temperature and velocity, ignoring tube metallurgy. This is flawed for seawater applications:

1. **Corrosion Products as Internal Fouling:** On copper alloy tubing, the tube material actively corrodes, generating internal scale resistance (R_f). Conversely, titanium exhibits near-zero metal loss ($< 0.001\text{ mm/year}$), completely eliminating the corrosion fouling component.
2. **Surface Energy and Roughness:** Titanium retains a smooth, glass-like surface, suppressing particulate sedimentation, whereas copper alloys develop high surface roughness that promotes scale nucleation.

3. **Velocity Limits:** TEMA bases its default figures on velocities of ≤ 1.5 m/s. Titanium permits operational velocities of 3–4 m/s, drastically lowering fouling deposition rates.

Consequently, specialized condenser standards (e.g., **HEI Standards for Steam Surface Condensers**) recommend reducing design fouling resistances by roughly 50% when replacing copper alloys with titanium ($0.00040 \rightarrow 0.00018\text{--}0.00020$ m²K/W).

2 Exchanger Architecture and Parameter Selection

- **TEMA Type AES Configuration:**

- *Front Head (Type A):* Removable bonnet cover allowing straight access for tubeside mechanical rod cleaning without dismantling seawater piping headers.
- *Shell (Type E):* Standard single-pass crossflow condensing shell.
- *Rear Head (Type S - Floating Head with Backing Device):* Accommodates severe differential thermal expansion between carbon steel shell and tubes subjected to high terminal thermal gradients ($\Delta T = 154.2^\circ\text{C} - 18.0^\circ\text{C} = 136.2^\circ\text{C}$).

- **Bundle Geometry:** 30° Triangular pitch provides maximal packaging density for clean condensing steam.
- **Baffling Scheme:** No-Tubes-In-Window (NTIW) segmental baffles with 25% diametral cut and 700 mm central spacing were implemented to suppress acoustic resonance and crossflow tube bundle vibration while keeping shell pressure drop well beneath the 25.0 kPa ceiling.

3 Thermal-Hydraulic Design & Metallurgical Evaluation

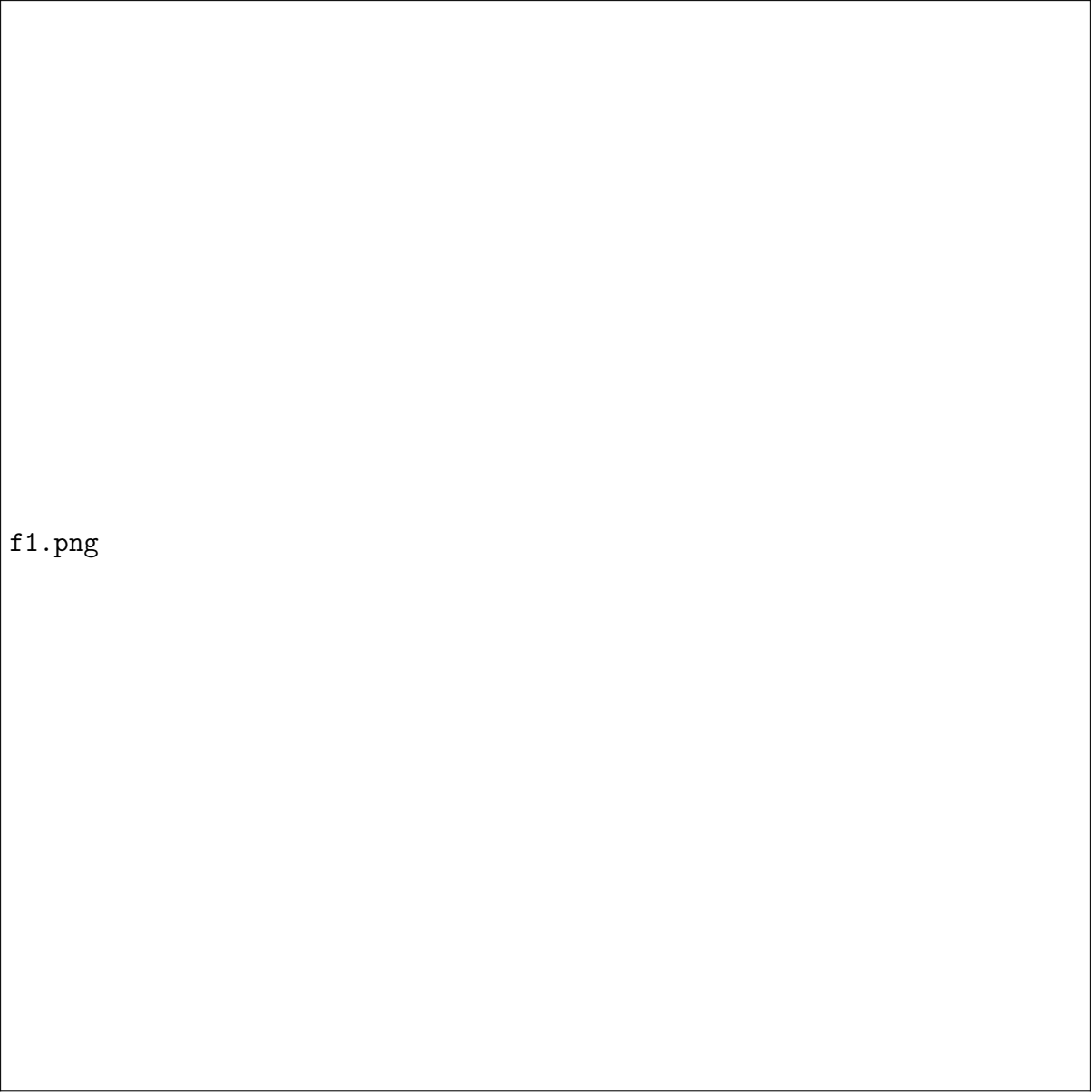
3.1 Material Comparison: Base Aluminum-Brass vs. Standard Titanium

The base exchanger was sized in HTRI **Design Mode** using Aluminum-Brass ($k \approx 100$ W/m · K, $t_w = 1.245$ mm). Subsequently, the shell envelope ($D_{\text{shell}} = 1168.4$ mm) was locked in **Rating Mode** to evaluate standard Titanium Grade 2 ($k \approx 21.9$ W/m · K, $t_w = 1.245$ mm).

Table 1: Performance comparison: Aluminum-Brass vs. Standard Titanium (Fixed Shell Diameter).

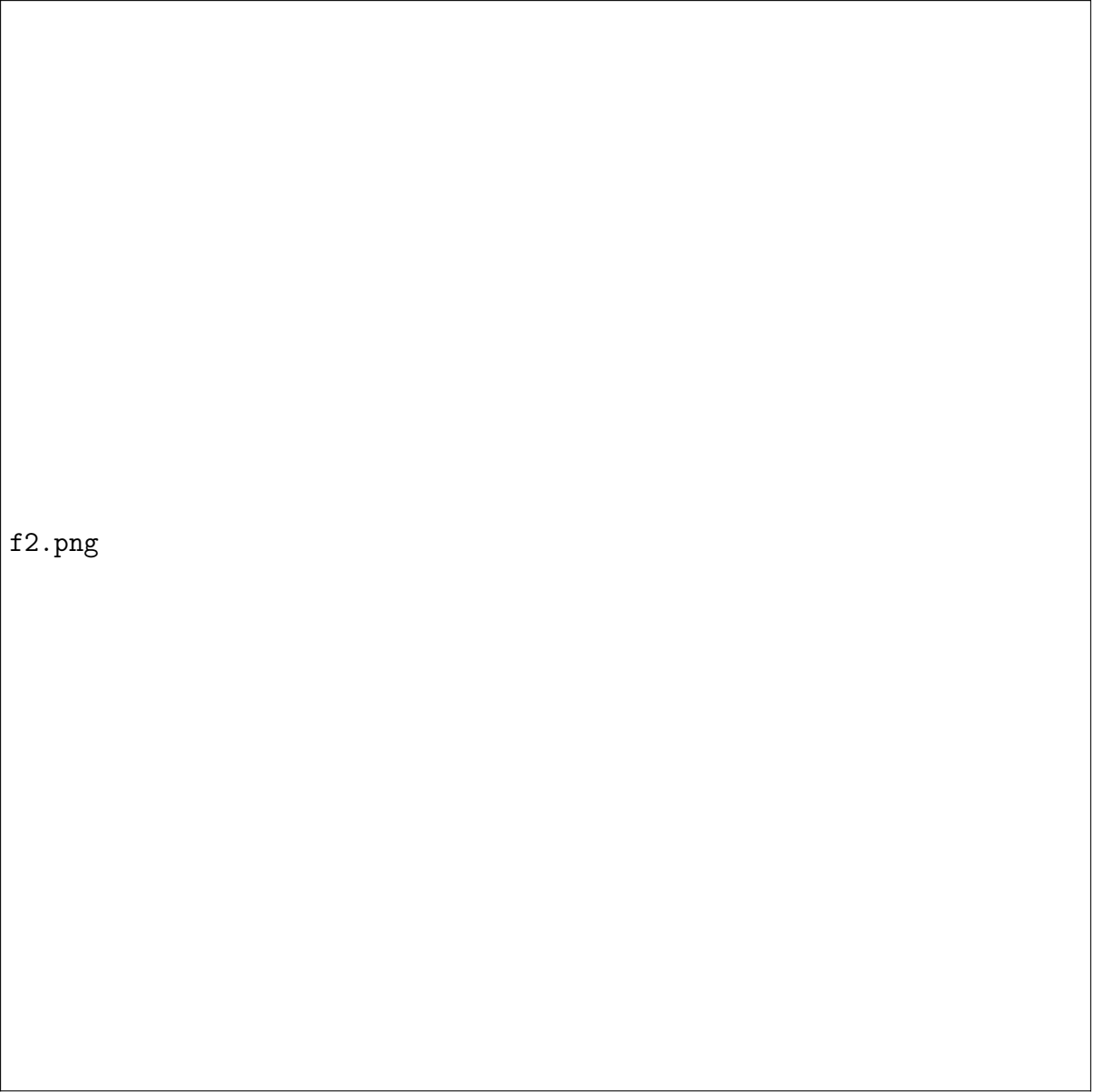
Design Parameter	Aluminum-Brass	Titanium (Grade 2)	Engineering Variation
Execution Mode	Design Mode	Rating Mode	Fixed Envelope
Shell Inside Diameter	1168.4 mm	1168.4 mm	Constant
Tube Metallurgy	Al-Brass (78Cu, 2Al)	Titanium Grade 2	Lower Conductivity
Tube Wall Thickness	1.245 mm	1.245 mm	Constant
Total Tube Count (N_t)	943	1184	+25.5% (+241 tubes)
Number of Tube Passes	2 passes	1 pass	Reconfigured
Overall Coefficient (U)	1531.2 W/m ² K	1210.3 W/m ² K	−21.0%
Tube Metal Resistance	1.97%	7.99%	$\approx 4\times$ increase
Effective Area (A)	131.68 m ²	166.23 m ²	+26.2%
Tubeside Pressure Drop	30.77 kPa	5.16 kPa	−83.2%
Shellside Pressure Drop	10.10 kPa	9.61 kPa	Constant (< 25 kPa)
Overdesign Margin	0.92%	0.90%	Target Matched

Engineering Discussion: Due to titanium’s low thermal conductivity (21.9 vs. 100 W/m·K), wall conductive resistance quadruples (1.97% $\rightarrow$ 7.99%). Delivering the 24.0 MW duty within identical shell dimensions requires adding 241 tubes, demonstrating that simple 1:1 metallurgy replacement imposes a severe capital penalty.



f1.png

Figure 1: HTRI TEMA Specification Sheet – Base Aluminum-Brass Design ($t_w = 1.245$ mm, 943 Tubes).



f2.png

Figure 2: HTRI TEMA Specification Sheet – Standard-Gauge Titanium Design ($t_w = 1.245$ mm, 1184 Tubes).

3.2 Optimization via Thin-Wall Titanium ($t_w = 0.559$ mm)

Titanium's high tensile and fatigue strength permits utilizing thin-wall tubing (0.559 mm ≈ 500 μ m) without risking hydraulic collapse.

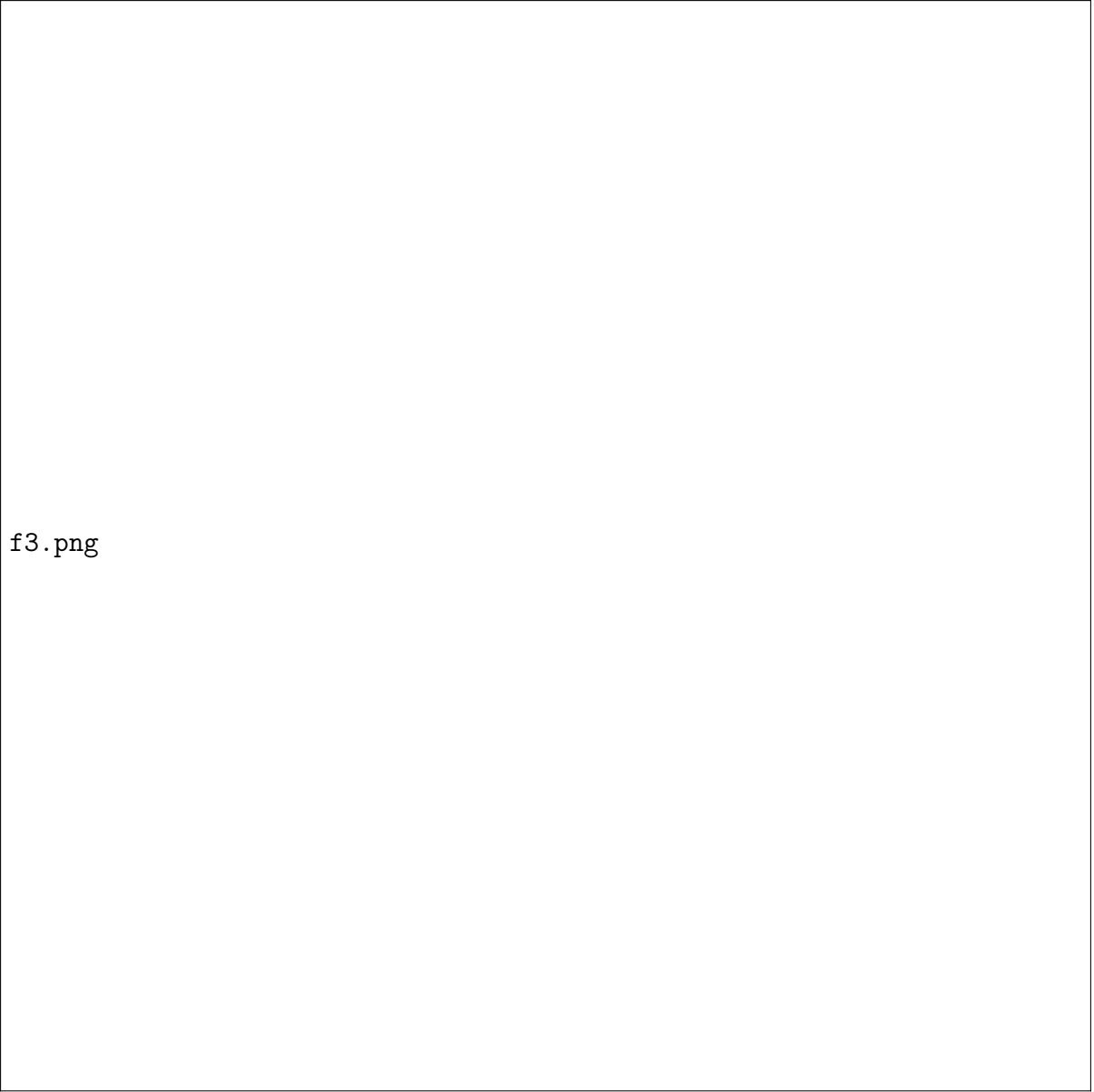
Table 2: Optimization impact: Standard Titanium vs. Thin-Wall Titanium.

Parameter	Standard Titanium	Thin-Wall Titanium	Net Benefit
Tube Thickness (t_w)	1.245 mm	0.559 mm	−55.1% reduction
Total Tube Count (N_t)	1184	1076	108 fewer tubes
Effective Surface Area	166.23 m ²	150.86 m ²	−9.2% reduction
Overall Coefficient (U)	1210.3 W/m ² K	1334.3 W/m ² K	+10.2% enhancement
Metal Wall Resistance	7.99%	3.81%	Halved bottleneck
Tubeside Pressure Drop	5.16 kPa	4.76 kPa	−7.7%
Tubeside Bulk Velocity	0.89 m/s	0.84 m/s	Safe low erosion

Quantitative Comparison vs. Base Al-Brass:

$$\Delta N = N_{\text{Thin-Ti}} - N_{\text{Al-Brass}} = 1076 - 943 = +133 \text{ tubes} \quad (+14.1\%) \quad (1)$$

Thinning the tube wall recovers more than half of the area penalty seen in standard-gauge titanium.



f3.png

Figure 3: HTRI TEMA Specification Sheet – Optimized Thin-Wall Titanium Design ($t_w = 0.559$ mm, 1076 Tubes).

4 HTRI Runtime Diagnostics and Vibration Analysis

During execution, four specific warning messages were diagnosed:

1. **Acoustic Vibration Warning:** HTRI flagged potential fundamental transverse acoustic resonance driven by high-velocity vapor vortex shedding. *Resolution:* HTRI recommended installing deresonating baffles at $0.45 \times D_{\text{shell}}$. This detunes the acoustic cavity natural frequency without impeding crossflow condensation.
2. **Thermal Conductivity Temperature Limit (Al-Brass):** Bulk condensing steam temperatures (154.2°C) exceeded the experimental database validation cap of 176.1°F (80°C) for material code 45. Thermal conductivity was clamped at the bound; resulting deviation in U was negligible ($< 0.2\%$).

3. **Rear Head Internal Diameter Limitation:** A geometric check on the Type S floating head indicated tight clearance between the floating tubesheet backing device and shell ID. This is addressed during structural mechanical detailing.
4. **ASME Section VIII Div. 1 Disclaimer:** Standard HTRI notice declaring vessel wall thicknesses and weights are preliminary estimations and require formal PVElite mechanical verification.

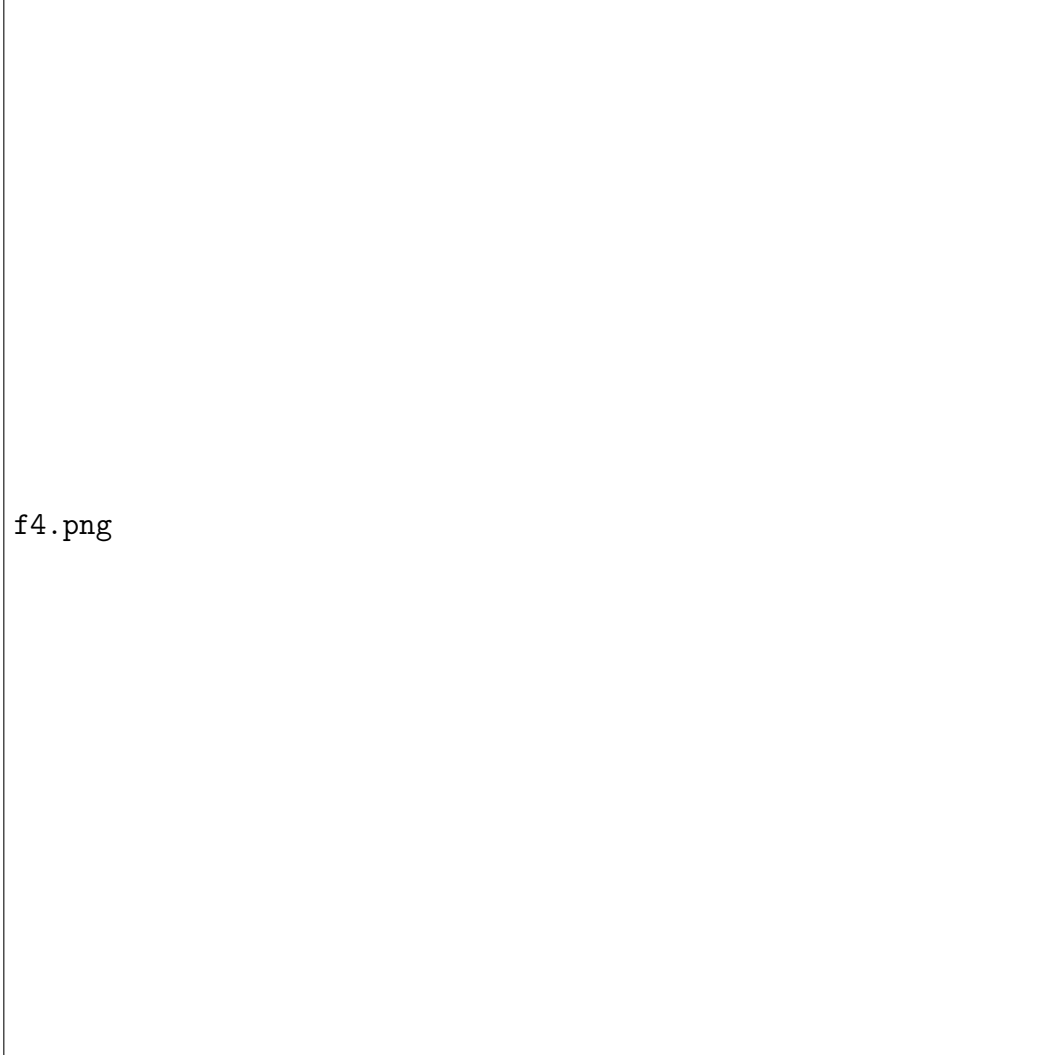


Figure 4: HTRI Xist Runtime Messages and Acoustic Vibration Diagnostics.

5 Analytical Validation via MATLAB and Root-Cause Analysis

An in-house 1D thermal resistance script was deployed to predict the required tube count when transitioning from Al-Brass to thin-wall Titanium:

$$\frac{1}{U_o} = R_{\text{wall}} + R_{\text{other}} = \frac{D_o \ln(D_o/D_i)}{2k} + \frac{D_o}{D_i} R_{\text{conv,in}} + R_{\text{conv,out}} + R_{\text{foul}} \quad (2)$$

Assuming R_{other} remains invariant and holding thermal duty constant:

$$N_{\text{est}} = N_{\text{base}} \times \left(\frac{U_{\text{base}}}{U_{\text{new}}} \right) \quad (3)$$

Table 3: MATLAB 1D analytical model vs. HTRI multi-segment rating.

Metric	Al-Brass (Base)	Titanium (MATLAB)	Titanium (HTRI Actual)
Overall Coefficient (U)	1531.20 W/m ² K	1497.49 W/m ² K	1334.30 W/m ² K
Required Tube Count (N_t)	943	965	1076
Model Error Relative to HTRI	—	−10.32%	Benchmark

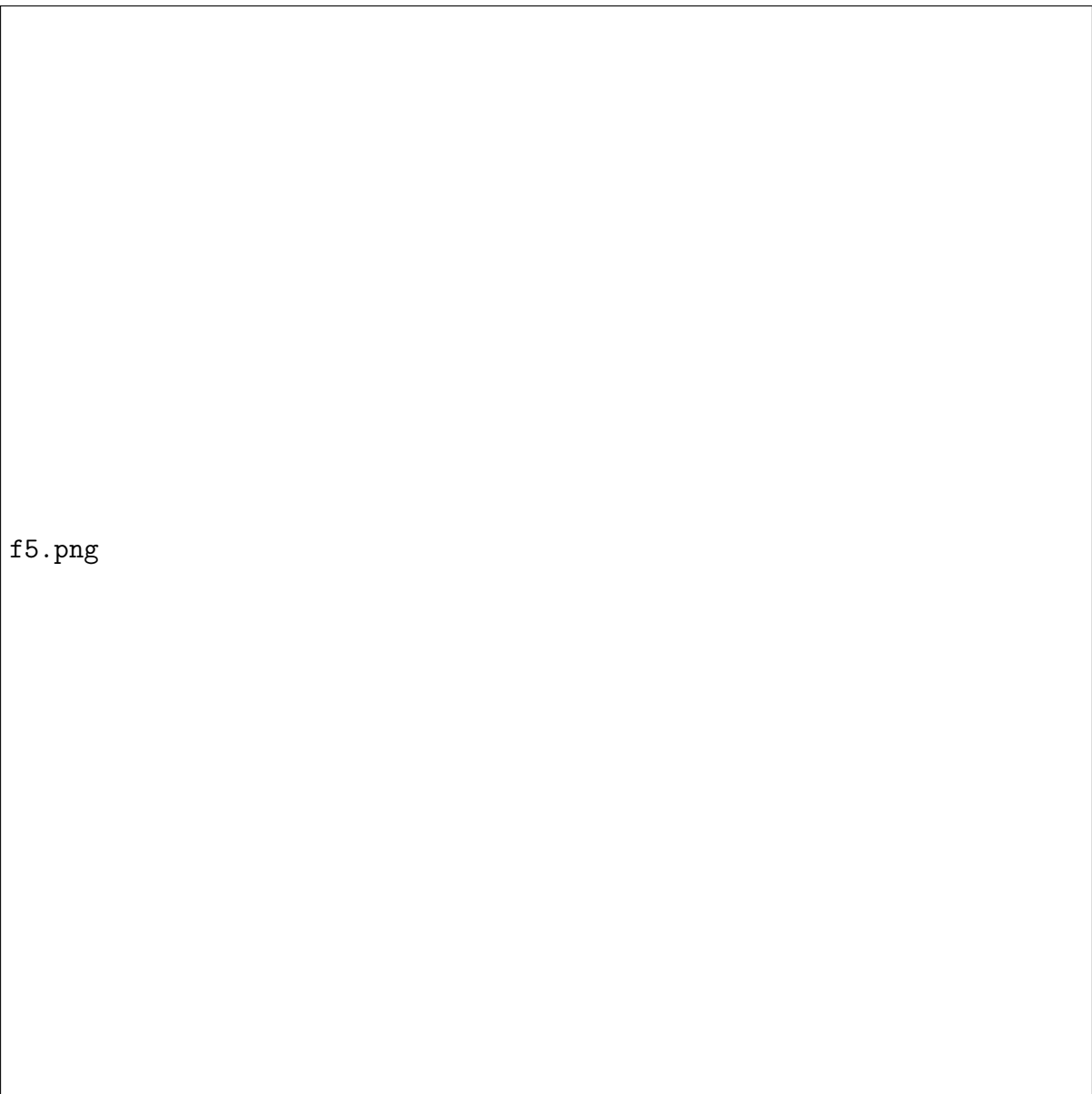


Figure 5: MATLAB Command Window Output: Analytical Sizing Discrepancy (Error = 10.32%).

5.1 Root-Cause Diagnosis of the 10.3% Underestimation

The MATLAB script underestimated tube count by 111 tubes due to coupled **thermal-hydraulic feedback**:

1. **Internal Diameter Expansion:** Thinning the tube wall from 1.245 mm to 0.559 mm under a fixed OD (19.05 mm) expanded the inner diameter:

$$D_{i,\text{Al-Brass}} = 16.56 \text{ mm} \longrightarrow D_{i,\text{Thin-Ti}} = 17.93 \text{ mm} \quad (4)$$

2. **Velocity Attenuation:** Flow cross-sectional area per tube expanded by 17.3%. With total mass throughput held constant, seawater bulk velocity dropped:

$$V = \frac{\dot{m}}{\rho A_{\text{flow}}} \implies V_{\text{Std-Ti}} = 0.89 \text{ m/s} \longrightarrow V_{\text{Thin-Ti}} = 0.84 \text{ m/s} \quad (5)$$

3. **Convective Heat Transfer Degradation:** Seawater forced convection obeys the Dittus-Boelter / Sieder-Tate correlation:

$$h_i \propto \frac{k}{D_i} Re^{0.8} \propto \frac{V^{0.8}}{D_i^{0.2}} \quad (6)$$

The drop in velocity combined with the expanded diameter degraded h_i .

Because the MATLAB script assumed R_{other} was constant, it overestimated U (1497.5 vs. 1334.3 W/m²K) yielding an unconservative tube count estimate. This demonstrates that tube gauge modifications cannot be treated as isolated conductive perturbations.

6 Summary and Conclusions

1. Substituting Aluminum-Brass with standard-gauge Titanium Grade 2 imposes a 25.5% tube count penalty (943 $\rightarrow$ 1184 tubes) due to titanium's lower conductivity (21.9 W/m · K).
2. Sizing with thin-wall titanium (0.559 mm) recovers 45% of this surface penalty, requiring 1076 tubes (+14.1% over base), cutting metal resistance to 3.81% while yielding complete immunity to seawater erosion-corrosion.
3. Coupling 1D conduction with internal hydraulics is critical; internal diameter enlargement attenuates velocity, degrading internal convection ($h_i \propto V^{0.8}$) by 10.3%.

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